# aquaponics_ORDERING

# AQUAPONICS controller: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: runs on BioTray; biological limits are placeholders. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

## 1. Order of work

1. Read the manual and the caveat above.
2. No custom circuit board: nothing to order from JLCPCB for this project.
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

## 3. Bought-in and mechanical items

## 1. Parts

BioTray board (see `biotray/MANUFACTURING_MANUAL.md`); DS18B20 waterproof probe; analog pH module + probe; analog EC module + probe; analog dissolved-oxygen module + probe; ADS1115 breakout on the I2C header; 12 V circulation pump; 12 V dosing (peristaltic) pump + pH-down reservoir; relay module for the aerator (the FAN output drives the relay coil, flyback is on the board); backup battery aerator (strongly recommended); GFCI-protected supply; sealed enclosure for the wet environment.

## 4. Confidence (read before you order)

- **Order confidence: Medium**: uses other boards plus bought-in modules; each module's pinout must be checked against its datasheet.
- **Works-as-intended confidence: Low-Medium**: Control and forecasting core passes 13 checks; biological limits are placeholders; no ammonia or nitrite sensing; live animals depend on it.
- **First thing to do:** Run a week with water only; tune limits with an aquaculture expert before any fish.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

## 5. Where to buy and whom to approach

| Item type | India (fast, GST invoice) | Overseas (cheaper, 1 to 3 weeks, customs) |
|---|---|---|
| Printed circuit board and SMD assembly | JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price) | **JLCPCB** (jlcpcb.com), PCBWay (pcbway.com) |

| Item type | India (fast, GST invoice) | Overseas (cheaper, 1 to 3 weeks, customs) |
|---|---|---|
| Connectors, headers, passives, ICs by LCSC number | element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts | **LCSC** (lcsc.com) with the same cart as JLCPCB |
| Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors | Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in | AliExpress, Seeed Studio, Waveshare |
| Motors, stepper drivers, wheels | Robu.in, Robocraze, local robotics shops | StepperOnline (omc-stepperonline.com) |
| Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers | Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice) | not recommended by post (shipping rules for batteries) |
| Laser-cut and 3D-printed parts | Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece | JLC3DP (jlc3dp.com), PCBWay 3D printing |
| Fasteners, tubing, hardware, enclosures | Local hardware, Robu.in, Amazon.in | n/a |

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.